THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS CHAPTER TEST SET A

Chapter 4: **Expressions Using Letter-Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If x = 4, the value of 3x + 5 is: 1

(a) 20 **(b)** 27 **(c)** 12 **(d)** 17

2. If x = 5, the value of 6x + 2 is: 1

(a) 42 **(b)** 38 **(c)** 13 **(d)** 32

3. Simplify 3x + 5x − 2x. 1

(a) $6x^2$ **(b)** 6x **(c)** 10x **(d)** 13x

4. Simplify 8x + 6x − 2x. 1

(a) $12x^2$ **(b)** 12x **(c)** 16x **(d)** 46x

5. ‘5 more than a number n’ is written as: 1

(a) 5 − n **(b)** n − 5 **(c)** 5n **(d)** n + 5

6. The perimeter of a square of side s is: 1

(a) s + 4 **(b)** 4s **(c)** 2s **(d)** s^2

7. In the pattern 5, 8, 11, 14 … the nth term is: 1

(a) n + 3 **(b)** 3n + 5 **(c)** 3n + 2 **(d)** 5n − 2

8. Which pair are like terms? 1

(a) 4a and 4b **(b)** 5 and 5y **(c)** 3x and $3x^2$ **(d)** 4a and 9a

9. The expression 2(x + 3) equals: 1

(a) 2x + 6 **(b)** 2x + 5 **(c)** x + 6 **(d)** 2x + 3

10. An expression with a single term is called a: 1

(a) trinomial **(b)** binomial

(c) polynomial of 3 terms **(d)** monomial

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is a letter-number? Why do we use it? 2

12. Write an expression: ‘7 less than 4 times a number x’. 2

13. Find the value of 5a − 2 for a = 6. 2

14. Simplify: 9y − 4y + 2y. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Simplify 6a + 4b − 2a + b − 3. 3

16. A taxi charges ₹20 plus ₹12 per km. Write an expression and find the fare for 7 km. 3

17. Find the nth term of 7, 12, 17, 22 … and the 20th term. 3

OR

Using x for the number of rows, write the number of chairs in a hall with 8 chairs per row and 5 extra chairs. 3

18. Evaluate $3a^2 - 2a + 1$ for a = 3. 3

OR

Simplify: 3(x + 2) + 2(x − 1). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Simplify 7a + 3b − 2a + 5b. 5
(ii) Evaluate it for a = 2, b = 3.
(iii) Write the expression for 4 times it.

OR

Case study: Matchsticks are used to make a row of triangles: 3 sticks for 1 triangle, 5 for 2, 7 for 3. 5
(i) Write the rule.
(ii) Sticks for 10 triangles.
(iii) How many triangles with 41 sticks?
(iv) Explain the rule.

20. (i) Expand 4(2x − 3). 5
(ii) Simplify 4(2x − 3) + 5x.
(iii) Evaluate for x = 2.

OR

(i) A rectangle has length (2x + 5) and width x. Write perimeter. 5
(ii) Write area.
(iii) Evaluate both for x = 3.

END OF QUESTION PAPER — Total: 40 Marks

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